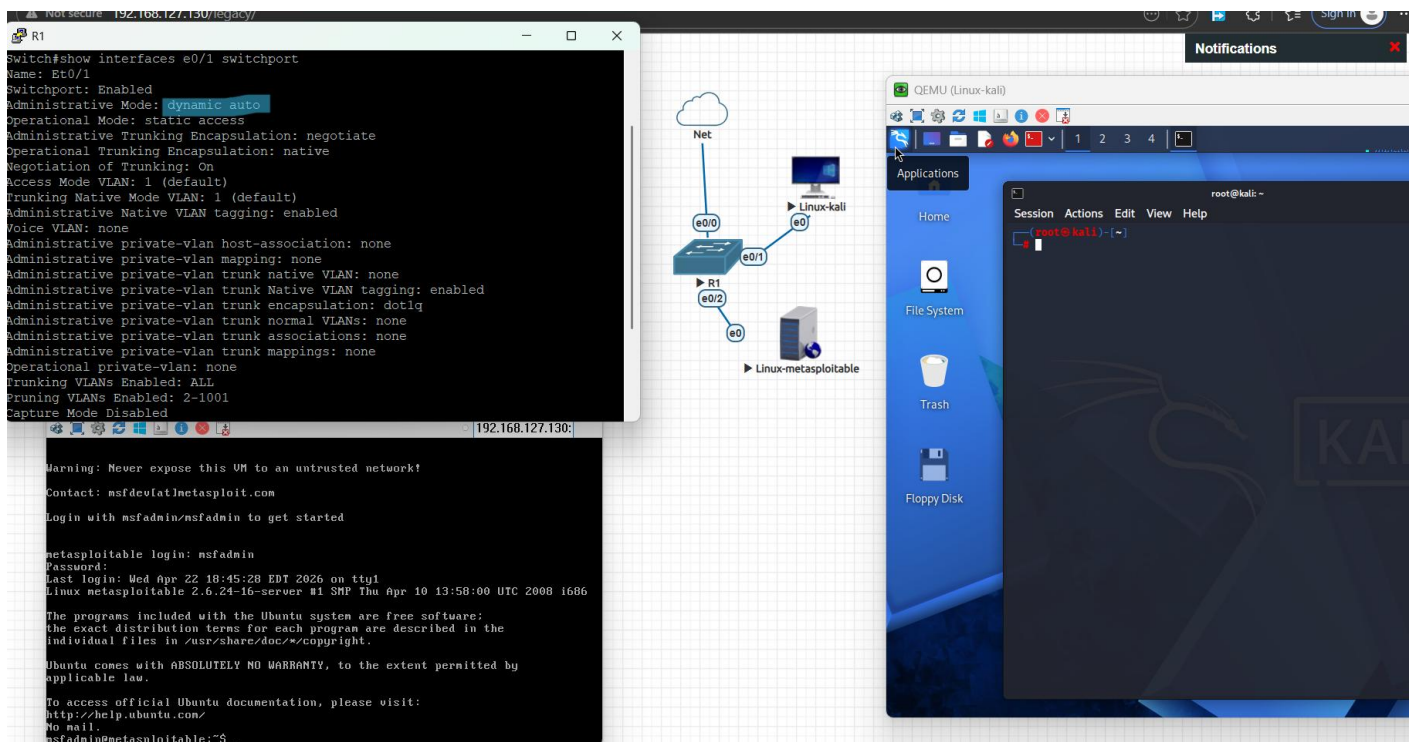


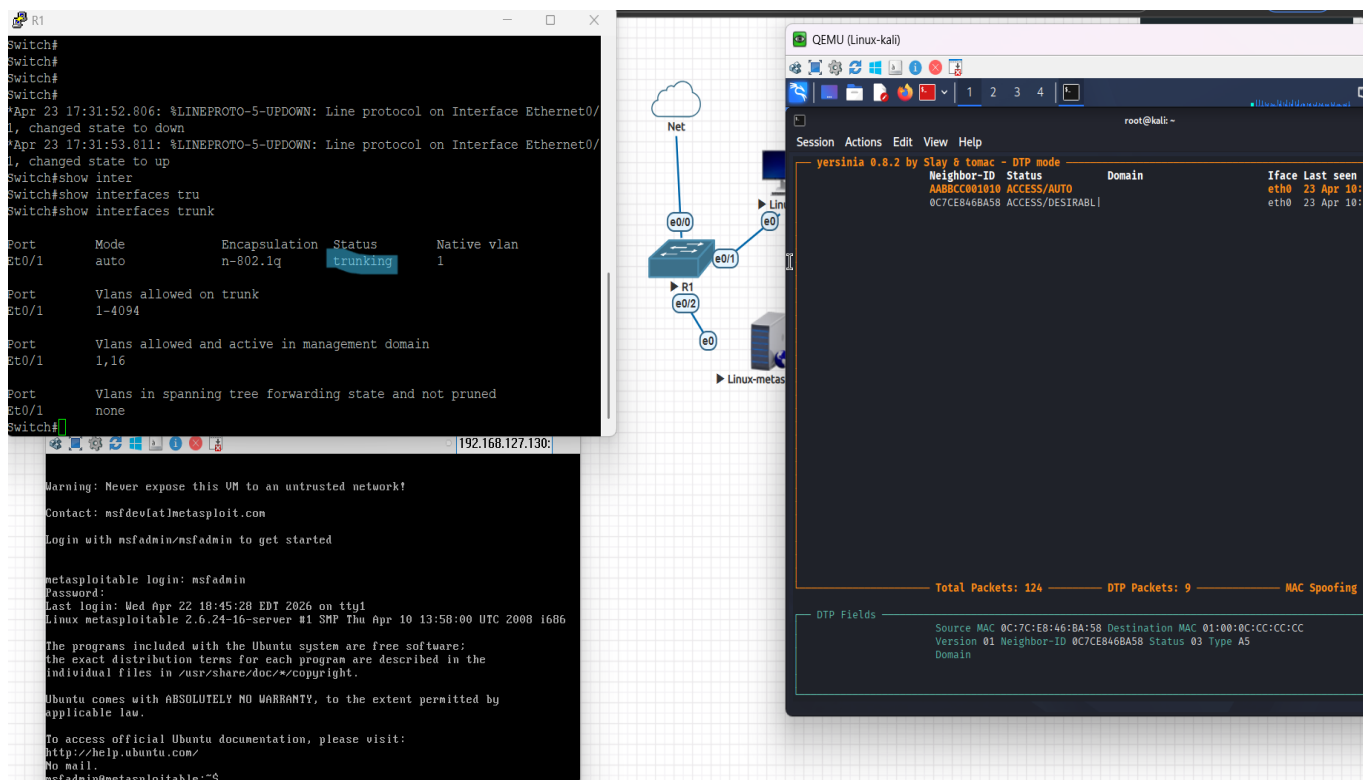
Layer 2 attacks

Introduction:

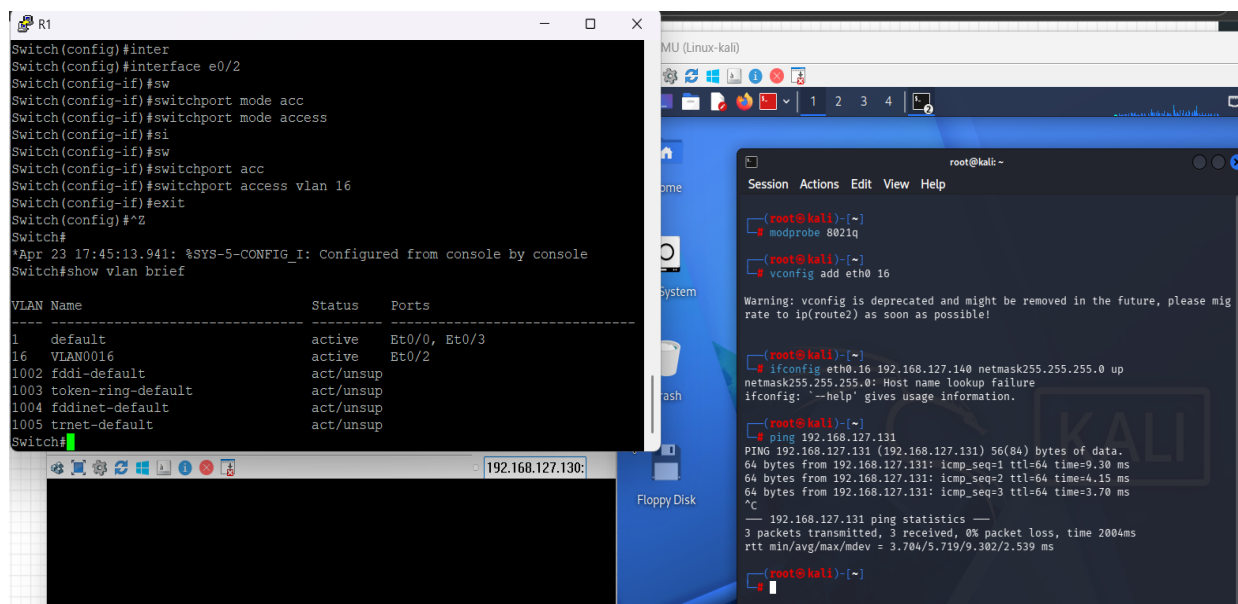
This lab simulates a Layer 2 security breach focused on **VLAN Hopping**. The primary objective is to demonstrate how an attacker can exploit the **Dynamic Trunking Protocol (DTP)** to bypass network segmentation. By masquerading as a legitimate switch (**Switch Spoofing**), we aim to establish a trunk link with the target switch, subsequently gaining unauthorized access to sensitive traffic within isolated VLANs (specifically VLAN 16).



I performed an initial scan of the switch port configuration using the command show interfaces Et0/1 switchport. The output reveals that the **Administrative Mode is set to dynamic auto**. This configuration is a security vulnerability, as it allows the port to automatically negotiate a trunk link if it receives DTP (Dynamic Trunking Protocol) frames from an external device.



In this stage, I utilized the Yersinia tool to launch a **Switch Spoofing** attack. By selecting the "Enabling Trunking" option, the Kali Linux machine began generating and sending malicious **DTP** (Dynamic Trunking Protocol) frames. These frames inform the switch that the connected device is a switch in desirable mode, forcing the target port to negotiate a **trunk link**.



In the final phase, I isolated the target device in VLAN 16 on the switch to simulate a restricted environment. On the Kali Linux side, I created a tagged sub-interface (eth0.16) and assigned it an IP address within the target's subnet. The successful Ping replies from 192.168.127.131 confirm a successful VLAN Hopping attack, proving that the network's logical segmentation was bypassed via the established trunk link.